

CS 101 - Algorithms & Programming I

Spring 2026 - Lab 9

Due: Week of May 4, 2026

*Remember the **honor code** for your programming assignments.*

*For all labs, your solutions must conform to the CS101 style **guidelines**!*

All data and results should be stored in variables (or constants where appropriate) with meaningful names.

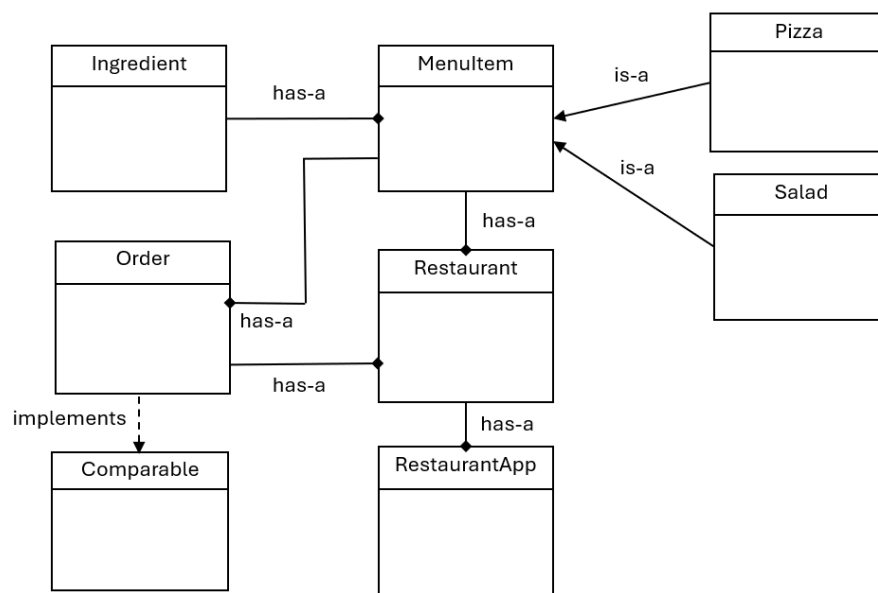
The objective of this lab is to understand and apply the object-oriented concepts, including abstract classes, interfaces, inheritance, and polymorphism, through an interactive restaurant management software. Before starting the implementation, you are encouraged to think about how the restaurant manager elements interact and organize them on paper as classes with clearly defined responsibilities and inheritance relations. Also, make sure where and how polymorphism is to be used.

0. Setup Workspace

Start VSC and open the previously created workspace named `labs_ws`. Now, under the `labs` folder, create a new folder named `lab9`. In this lab, you are to create Java classes/files (under `labs/lab9` folder) as described below. You are expected to submit the files for your initial solution as shown in the UML class diagram. For the revision, you may submit additional files, adding the "Rev" suffix to each revision file (e.g., `RestaurantRev`). **Please submit the files without compressing them, and ensure that no other or previous lab solutions are included.**

1. Overview

The goal of this lab is to complete a small restaurant ordering system. The classes in the starter code file represent different parts of the system. The application uses inheritance, abstraction, interfaces and interaction between classes. The basic UML diagram below indicates the relationships between the classes.



Download the starter_code file and extract. Carefully review the classes given to understand the given functionality and how the system is designed to work as a whole.

Do not modify the existing method signatures, class names, field names, or provided logic. The starter code has been structured intentionally, and changing existing functionality may cause other parts of the program to stop working.

Complete the assignment by writing code only in the sections marked by comments, especially the TODO areas. Avoid duplicating code or rewriting working functionality. Your goal is to complete the missing parts of the system while keeping the original structure intact.

2. Sample Run

Sample run data is shown below, and you may use it to determine the expected outputs, format and behavior of the program. If a scenario is not shown, you may decide what should be displayed, as long as the output is clear, reasonable, and consistent with the rest of the program.

```
=== Restaurant Menu ===
```

1. View Menu
2. View Vegetarian Dishes
3. View Allergen-Free Dishes
4. Add Item to Order
5. Customize Order Item
6. Display Orders
7. Quit

```
Choose: 1
```

```
Menu:
```

1	Veggie Pizza	38.00
2	Cheese Pizza	35.00
3	Meat Pizza	38.00
4	Chicken Pizza	36.50
5	Garden Salad	21.50
6	Simple Salad	19.50

```
=== Restaurant Menu ===
```

1. View Menu
2. View Vegetarian Dishes
3. View Allergen-Free Dishes
4. Add Item to Order
5. Customize Order Item
6. Display Orders
7. Quit

```
Choose: 2
```

```
Vegetarian Menu:
```

Veggie Pizza	38.00
Cheese Pizza	35.00
Garden Salad	21.50
Simple Salad	19.50

=== Restaurant Menu ===

1. View Menu
2. View Vegetarian Dishes
3. View Allergen-Free Dishes
4. Add Item to Order
5. Customize Order Item
6. Display Orders
7. Quit

Choose: 3

Allergen Free Menu:

Veggie Pizza	38.00
Chicken Pizza	36.50
Garden Salad	21.50
Simple Salad	19.50

=== Restaurant Menu ===

1. View Menu
2. View Vegetarian Dishes
3. View Allergen-Free Dishes
4. Add Item to Order
5. Customize Order Item
6. Display Orders
7. Quit

Choose: 4

Menu:

1	Veggie Pizza	38.00
2	Cheese Pizza	35.00
3	Meat Pizza	38.00
4	Chicken Pizza	36.50
5	Garden Salad	21.50
6	Simple Salad	19.50

Enter choice:1

Enter order number (0 for new order): 0

Item added to order 1000.

=== Restaurant Menu ===

1. View Menu
2. View Vegetarian Dishes
3. View Allergen-Free Dishes
4. Add Item to Order
5. Customize Order Item
6. Display Orders
7. Quit

Choose: 4

Menu:

1	Veggie Pizza	38.00
2	Cheese Pizza	35.00
3	Meat Pizza	38.00
4	Chicken Pizza	36.50
5	Garden Salad	21.50
6	Simple Salad	19.50

Enter choice:5

Enter order number (0 for new order): 1000

Item added to order 1000.

=== Restaurant Menu ===

1. View Menu
2. View Vegetarian Dishes
3. View Allergen-Free Dishes
4. Add Item to Order
5. Customize Order Item
6. Display Orders
7. Quit

Choose: 4

Menu:

1	Veggie Pizza	38.00
2	Cheese Pizza	35.00
3	Meat Pizza	38.00
4	Chicken Pizza	36.50
5	Garden Salad	21.50
6	Simple Salad	19.50

Enter choice:4

Enter order number (0 for new order): 0

Item added to order 1001.

=== Restaurant Menu ===

1. View Menu
2. View Vegetarian Dishes
3. View Allergen-Free Dishes
4. Add Item to Order
5. Customize Order Item
6. Display Orders
7. Quit

Choose: 6

Order Number: 1001

Chicken Pizza

Order Total: 36.5

Order Number: 1000

Veggie Pizza

Garden Salad

Order Total: 59.5

=== Restaurant Menu ===

1. View Menu
2. View Vegetarian Dishes
3. View Allergen-Free Dishes
4. Add Item to Order
5. Customize Order Item
6. Display Orders
7. Quit

Choose: 5

Enter order number to customize: 1000

1 Veggie Pizza

2 Garden Salad

Enter item to customize: 1

Veggie Pizza

1. Add Ingredient
2. Remove Ingredient
3. Done

Enter choice: 1

- Dough
- Tomato
- Mushroom
- Cheese
- Beef
- Chicken
- Lettuce
- Olive

Enter ingredient to add: cheese

Added.

1. Add Ingredient
2. Remove Ingredient
3. Done

Enter choice: 2

Ingredients:

- Dough
- Tomato
- Mushroom
- Olive
- Cheese

Enter ingredient to remove: tomato

tomato removed from Veggie Pizza

1. Add Ingredient
2. Remove Ingredient
3. Done

Enter choice: 2

Ingredients:

- Dough
- Mushroom
- Olive
- Cheese

Enter ingredient to remove: beef

Veggie Pizza does not contain beef

1. Add Ingredient
2. Remove Ingredient
3. Done

Enter choice: 3

Quitting

=== Restaurant Menu ===

1. View Menu
2. View Vegetarian Dishes
3. View Allergen-Free Dishes
4. Add Item to Order
5. Customize Order Item
6. Display Orders
7. Quit

Choose: 7

Goodbye.